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Mercury's Mirror: A Flat-Spacetime Null-Shell Derivation of Perihelion Precession

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A flat-spacetime, time-symmetric null-contact framework is used to derive Mercury's perihelion precession without invoking spacetime curvature. Starting from a two-body, retarded-plus-advanced causal interaction, we show that the $\mathcal{O}(v^2/c^2)$ corrections assemble into a Darwin-type effective Lagrangian for the Sun–Mercury system. The resulting Binet equation

$$u'' + u = \frac{GM}{h^2} \left(1 + 3 \frac{GM}{c^2} u \right)$$

yields the secular advance

$$\Delta\varpi = \frac{6\pi GM}{a(1-e^2)c^2},$$

reproducing the observed $43''/\text{century}$ for Mercury. The same null-shell geometry accounts for the factor of two between Newtonian and relativistic light bending. These results support the claim that the standard 6π correction emerges from causal structure in Minkowski space, with curvature serving as an effective description rather than a necessary ontology.

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1. Introduction

Einstein's original account of Mercury's perihelion advance is widely taught as a clean prediction of spacetime curvature. Here a different route is taken: a flat-spacetime, time-symmetric null-contact model. The method is conservative in dynamics but unconventional in interpretation. It keeps interactions strictly on null separations and treats the retarded and advanced contributions symmetrically. At $\mathcal{O}(v^2/c^2)$ this produces the familiar 6π shift and the full light-deflection value, suggesting that the observed corrections follow from causal geometry rather than metric curvature.

2. Framework: local null contact and time symmetry

Two axioms are adopted.

Local null contact. Interaction occurs only at null separation. A test body's force is determined by source events on its past and future light cones.

Time symmetry. Fundamental dynamics are invariant under time reversal, implemented as the average of retarded and advanced contributions.¹ For a Sun–Mercury system,

$$\mathbf{F}(t) = -\frac{GMm}{2r^2} \left[\hat{\mathbf{r}}_{\text{ret}}(t) + \hat{\mathbf{r}}_{\text{adv}}(t) \right], \quad (1)$$

with $\hat{\mathbf{r}}_{\text{ret/adv}}$ pointing from Mercury's instantaneous position to the Sun's null-displaced positions at times $t \mp r/c$ along the Sun's worldline. Expanding the time-symmetric two-body action to $\mathcal{O}(v^2/c^2)$ yields an effective Darwin-type Lagrangian.^{2,3}

3. Perihelion precession

3.1. *Darwin-type Lagrangian at $\mathcal{O}(v^2/c^2)$*

For relative motion with reduced mass μ and separation r , the retarded/advanced angular displacement generates velocity-dependent terms. The time-symmetric average gives

$$L = \frac{1}{2}\mu(\dot{r}^2 + r^2\dot{\theta}^2) + \frac{GM\mu}{r} + \frac{GM\mu}{2c^2r} \left[3(\dot{r}^2 + r^2\dot{\theta}^2) - \dot{r}^2 - \frac{GM}{r} \right], \quad (2)$$

where the cross terms odd in time reverse cancel in the retarded-plus-advanced average.¹

3.2. *From L to the Binet equation*

Define the specific angular momentum $h = r^2\dot{\theta}$, which is constant to this order. With $u(\theta) = 1/r$ and

$$\dot{r} = -\frac{h}{r} u', \quad \dot{r}^2 + r^2\dot{\theta}^2 = h^2(u'^2 + u^2),$$

the Euler–Lagrange equation for r , keeping only first-order ($1/c^2$) terms and eliminating t in favor of θ , reduces to

$$u'' + u = \frac{GM}{h^2} \left(1 + 3\frac{GM}{c^2} u \right). \quad (3)$$

The coefficient 3 traces to the combination $3(\dot{r}^2 + r^2\dot{\theta}^2) - \dot{r}^2$ in Eq. (2).

3.3. *Secular advance*

Solve Eq. (3) perturbatively, $u = u_0 + \epsilon u_1$, with $u_0 = (GM/h^2) [1 + e \cos \theta]$. The $\mathcal{O}(\epsilon)$ driving term shifts the oscillator frequency and yields, per orbit,

$$\Delta\varpi = \frac{6\pi GM}{a(1 - e^2)c^2}, \quad (4)$$

using $h^2 = GMa(1 - e^2)$. For Mercury this reproduces 43 arcseconds per century.

4. Consistency check: light deflection

The same null-contact kinematics give the full $4GM/(bc^2)$ light-bending value for impact parameter b . Newton's $2GM/(bc^2)$ appears when only one null-side contribution is counted; the time-symmetric prescription doubles it for null trajectories, matching the standard result.

5. Discussion

The derivation shows that the 6π perihelion correction and the factor-of-two in light bending arise from geometrical features of null-shell contact in flat spacetime. The curvature account can be viewed as an effective field representation of the same causal structure. Observable predictions coincide at this order; the interpretive difference sits in ontology rather than kinematics. Higher-order tests and two-body back-reaction can be handled within the same time-symmetric formalism.

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